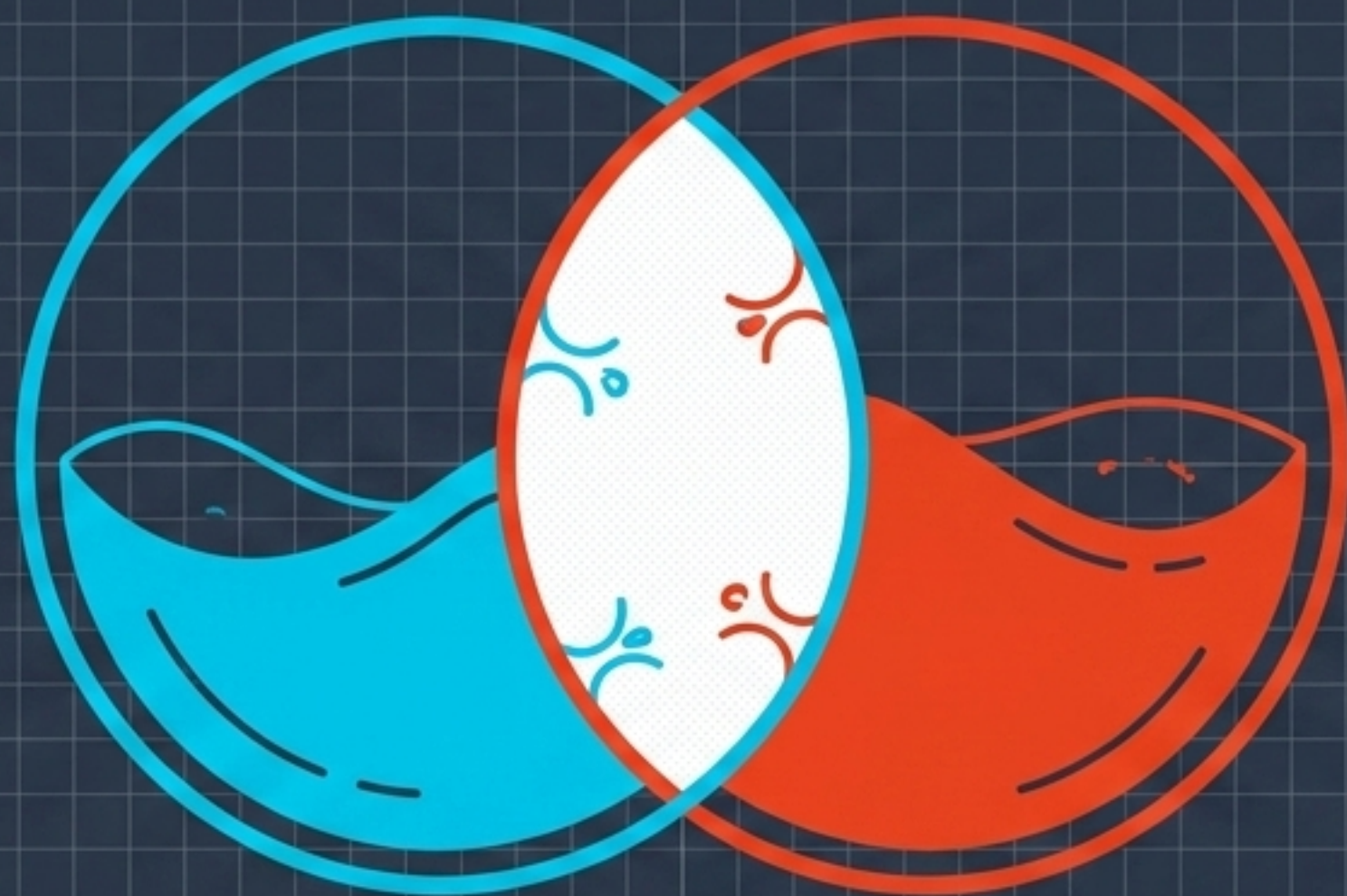




Two Fluids, Shared Pores

Rel-perm and wettability govern multiphase flow.

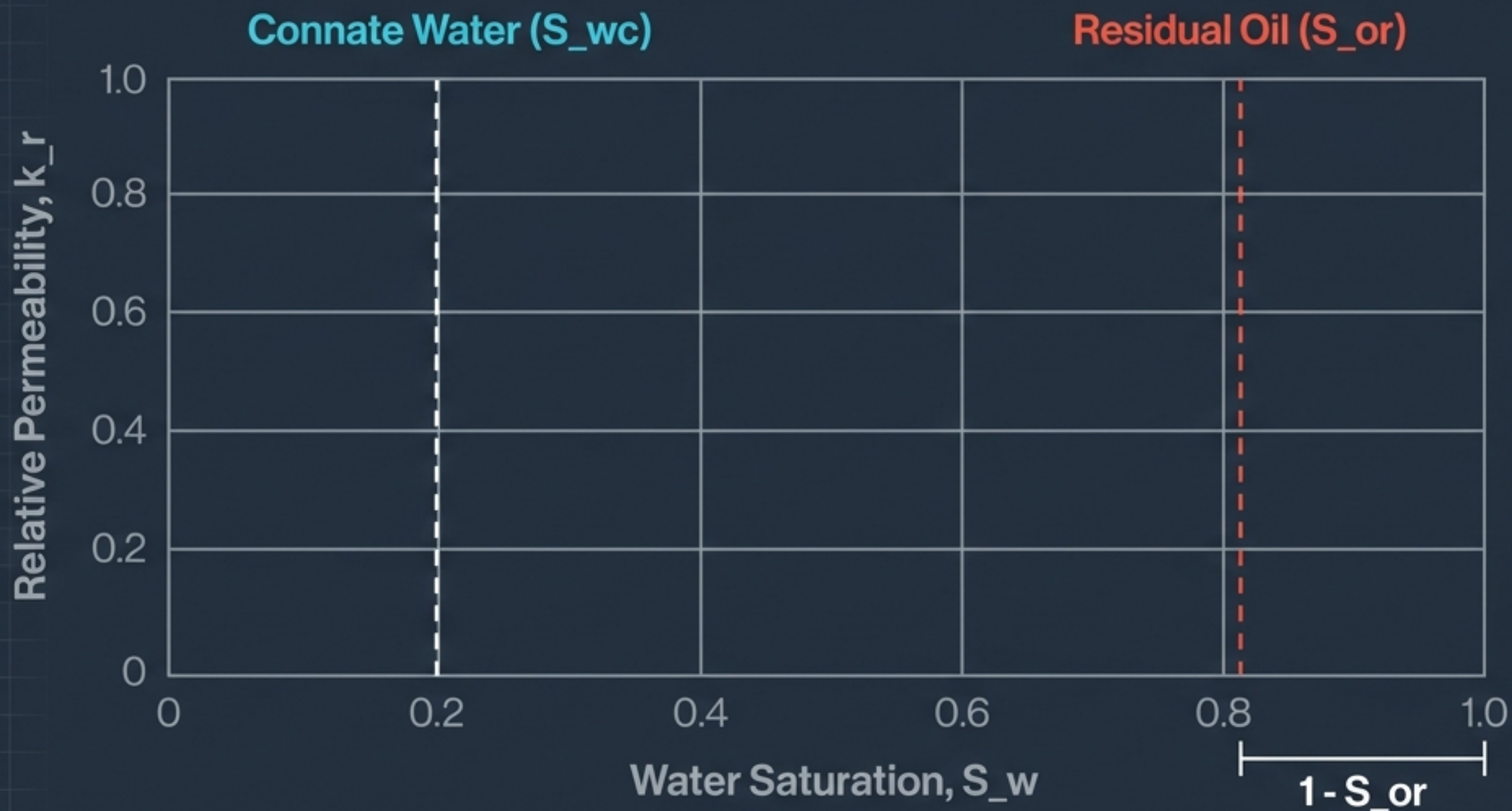


Presented by
Sajja K.Hano



Defining the boundaries of fluid saturation

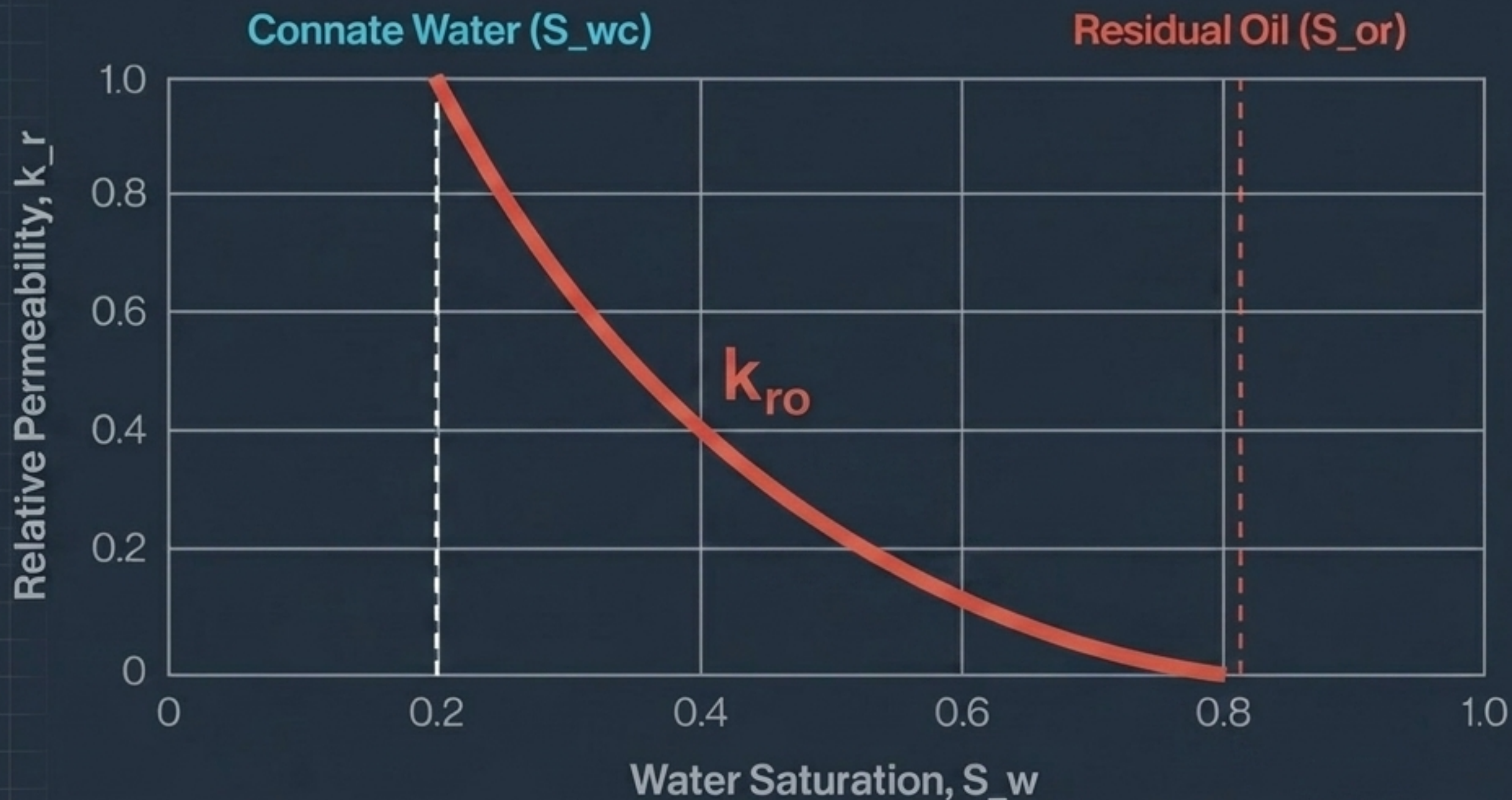
SL_TECH_NOTE_04



Flow only occurs between the limits of irreducible water and residual oil.

The decline of oil mobility

SL_TECH_NOTE_04

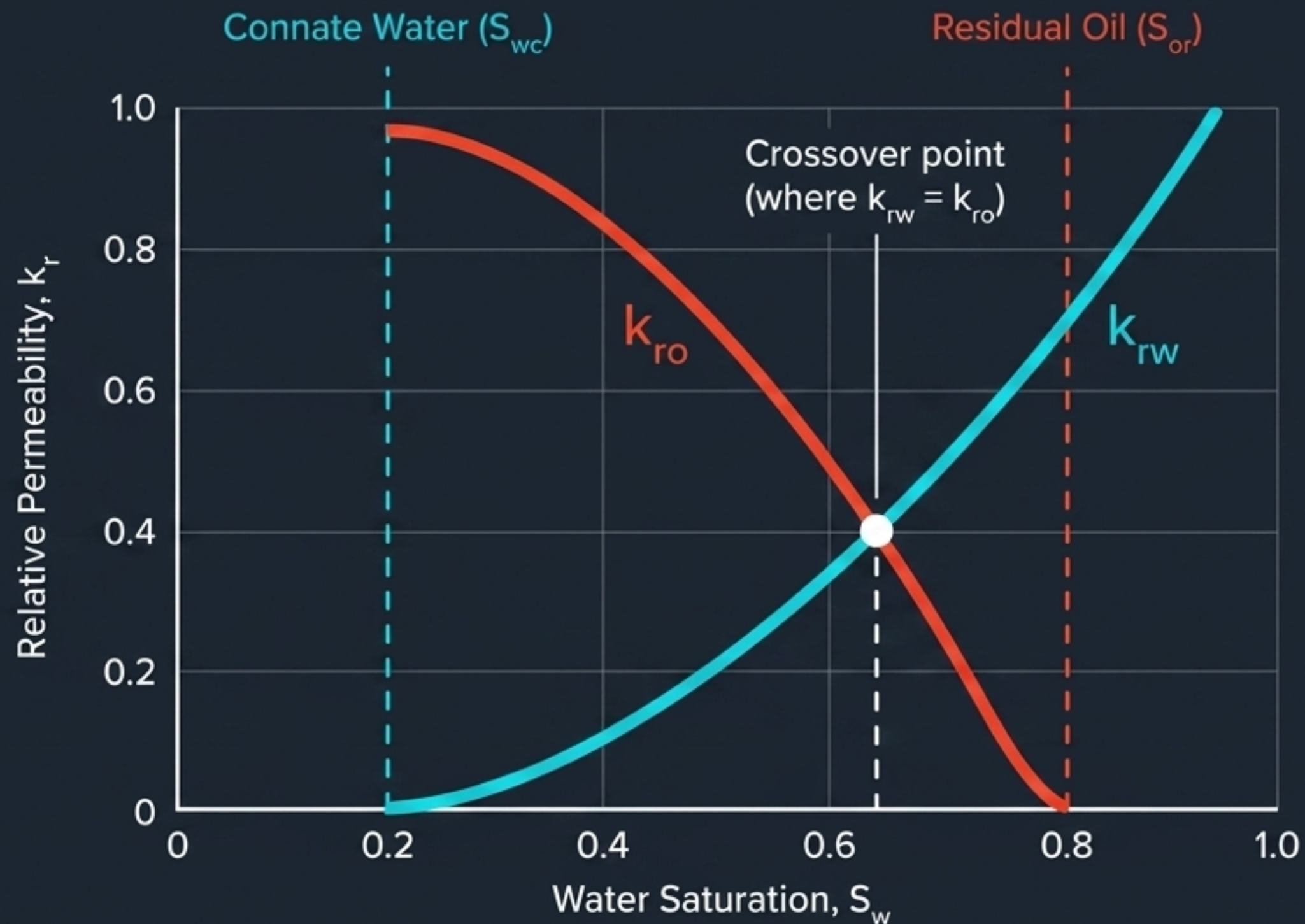


k_{ro} falls as S_w increases.

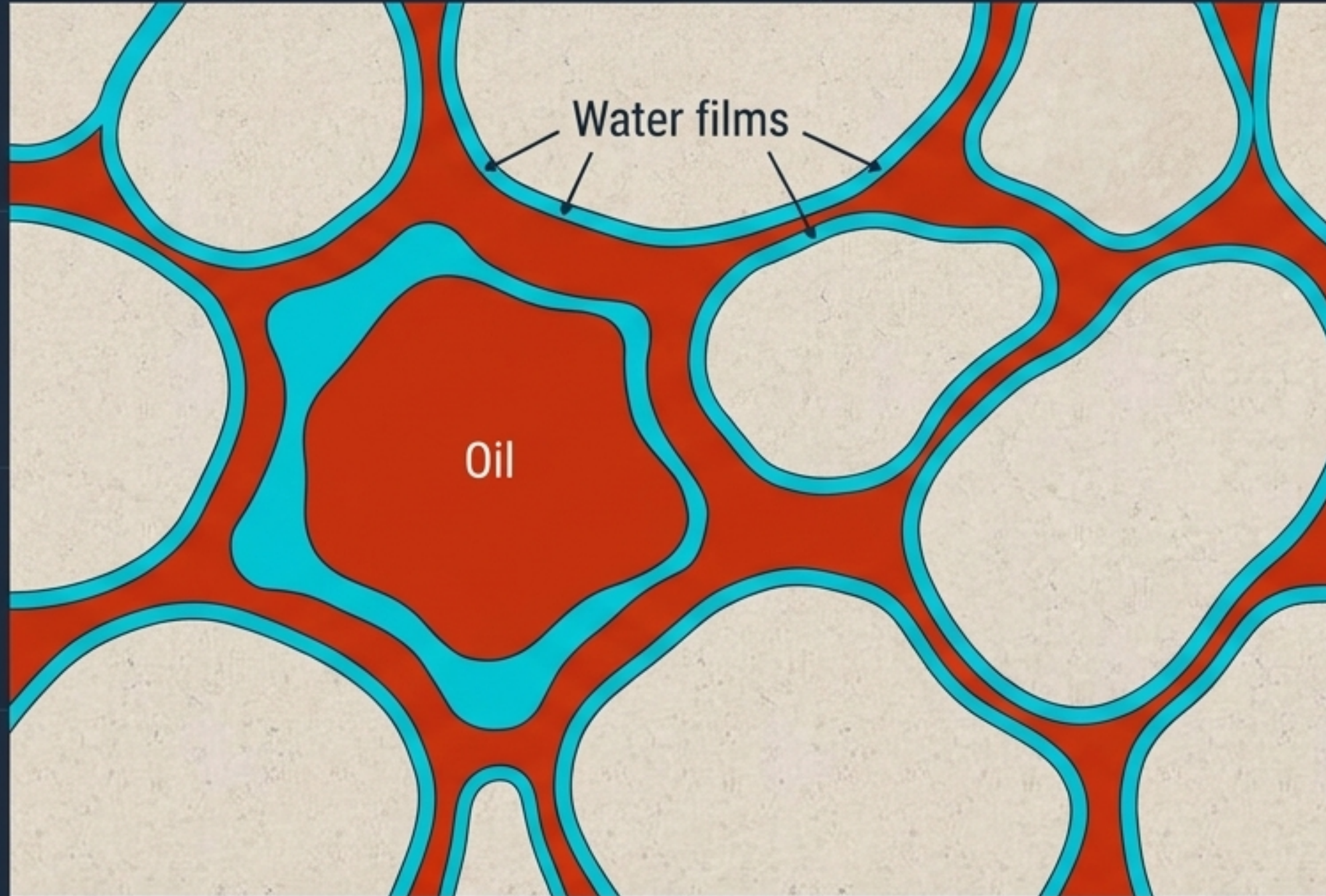
As water saturates the pore space, the relative permeability of oil systematically drops until it reaches zero at residual oil (S_{or}).

The rising water phase and the critical crossover

SL_TECH_NOTE_04



Pore scale physics in a water-wet system



State:

Water-Wet

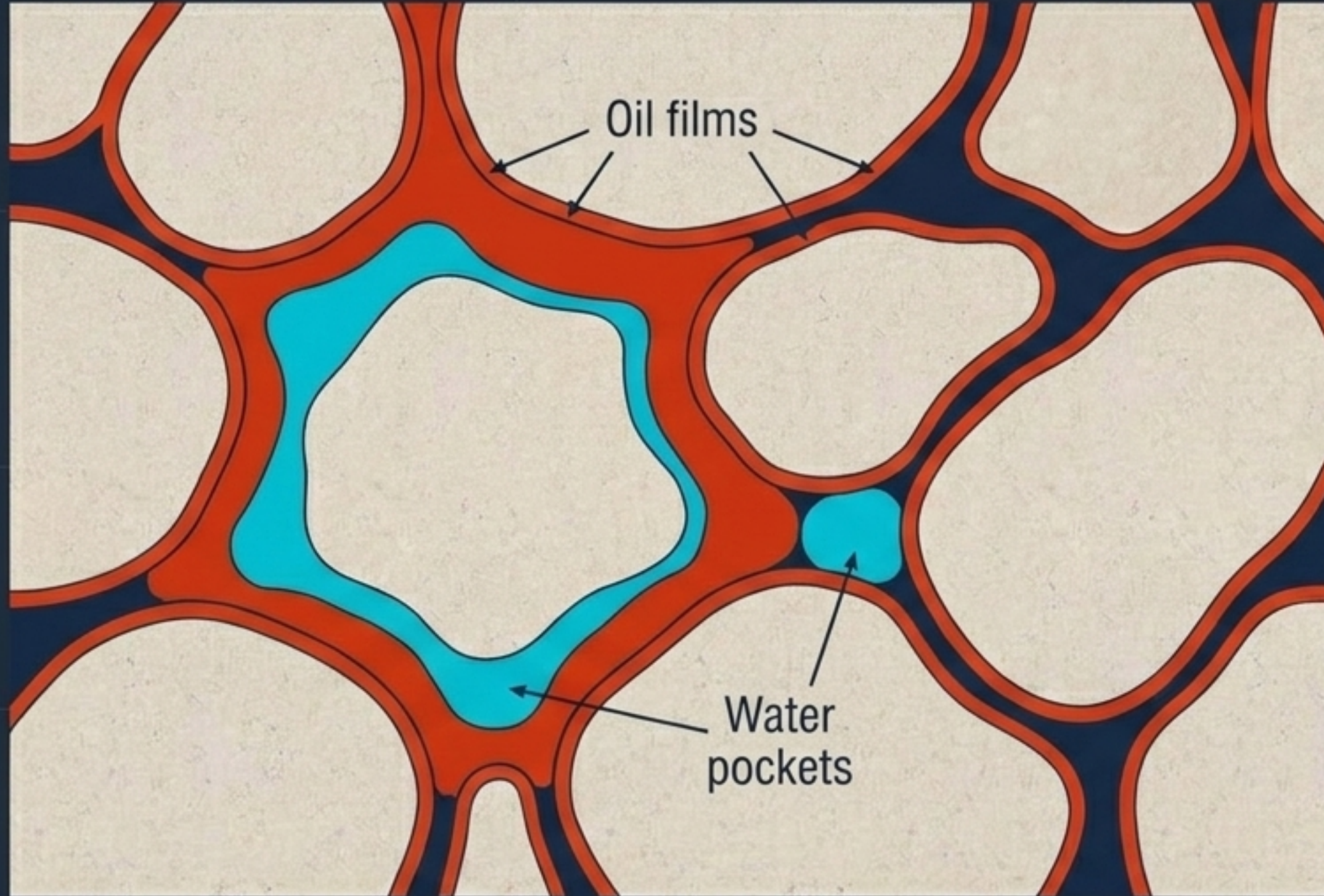
Behavior:

Water preferentially coats the rock surface, creating conductive films.

Result:

Oil is forced into the center of the pore network, allowing it to flow more freely until it is eventually snapped off and trapped.

The disruption of an oil-wet system



State:

Oil-Wet

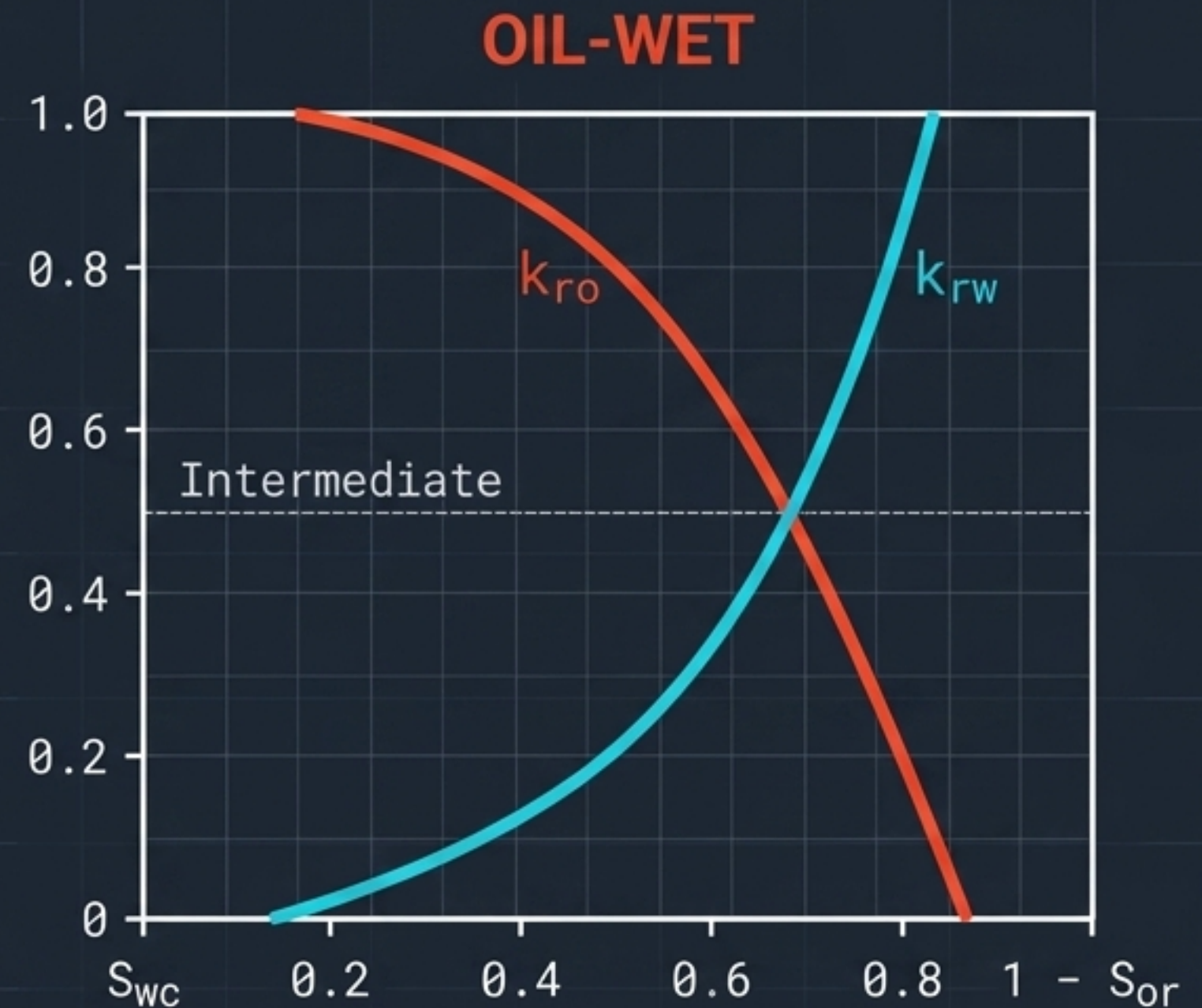
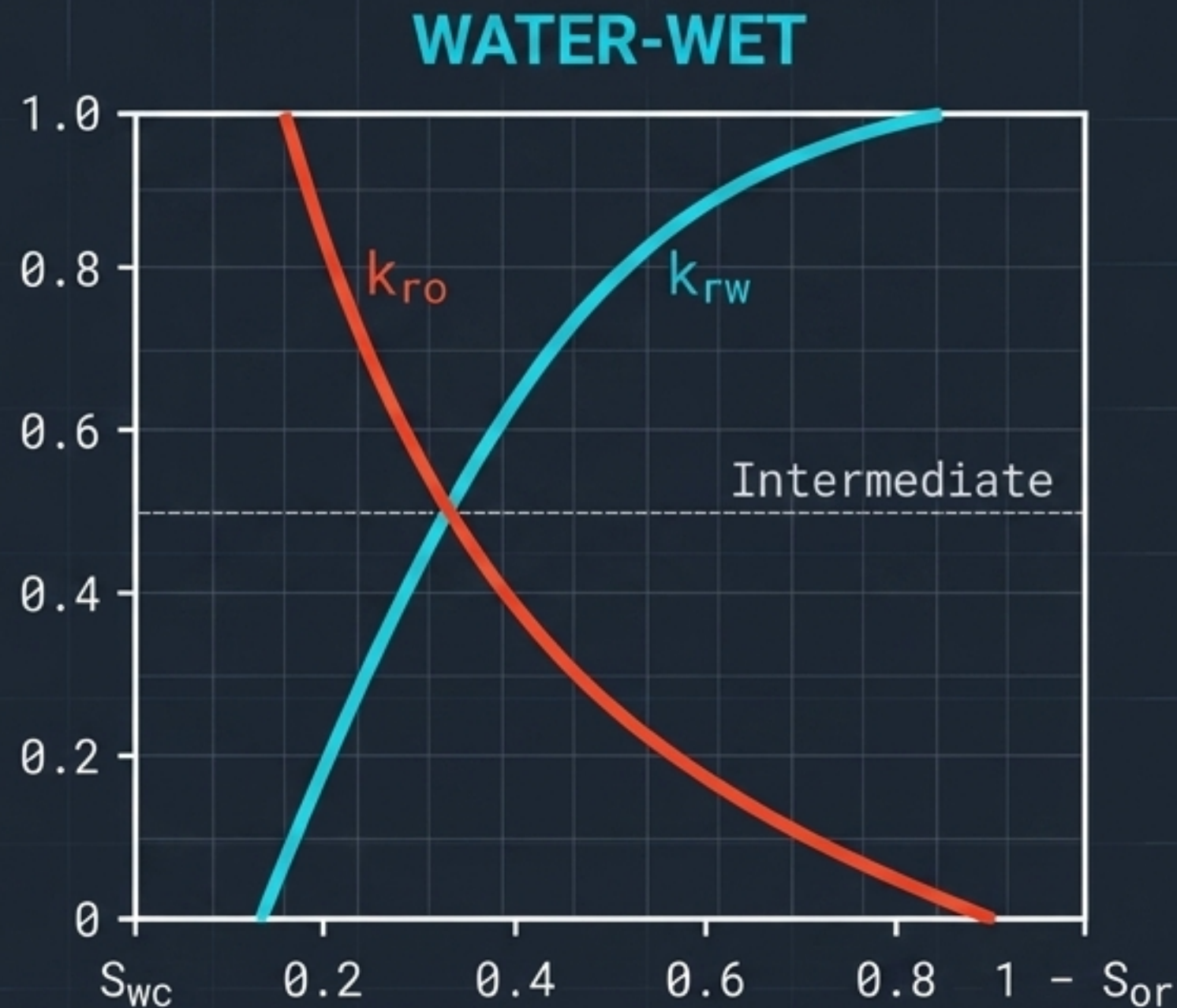
Behavior:

Oil preferentially coats the rock surface.

Result:

Water is forced into isolated pockets and channels. The oil clings to the rock, making it drastically harder to displace during a waterflood.

How wettability shifts the curves and crossover points



Diagnosing wettability from relative permeability data

MORE WATER-WET	MORE OIL-WET
 S_{or} (lower residual oil)	 S_{or} (higher residual oil)
 k_{rw} at given S_w	 k_{rw} at given S_w
 Crossover at lower S_w	 Crossover at higher S_w

Sourcing the data through Special Core Analysis



Special Core Analysis (SCAL)

Physical extraction of rock samples from the target formation.



Steady-state and Unsteady-state Tests

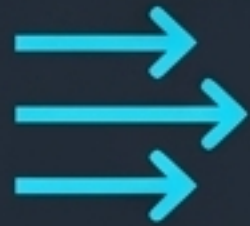
Dynamic fluid injection to measure pressure drops and fluid displacement in real-time.



Calibrated to the Reservoir

Lab data is mathematically corrected for in-situ pressure, temperature, and fluid properties.

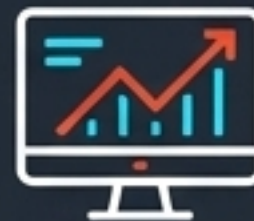
The commercial imperative of multiphase flow



Drives fractional flow and sweep efficiency during waterflooding operations.



Sets the physical limit of residual oil (S_{or}), defining the maximum possible recovery factor.



Serves as the key, foundational simulation input required for accurate reservoir modeling and financial forecasting.